

SHETH LUJ COLLEGE

Aim:-Naive Bayes' Classifier

- Implement the Naive Bayes' algorithm for classification.
- Train a Naive Bayes' model using a given dataset and calculate class probabilities.
- Evaluate the accuracy of the model on test data and analyze the results.

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IDLE Shell 3.13.7
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Daniyal khan/AI_PRACTICAL/naive.py =====
=====
Class Probabilities:
[0.33333333 0.34166667 0.325   ]

Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

Actual Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

Accuracy: 100.0 %

Confusion Matrix:
[[10 0 0]
 [0 9 0]
 [0 0 11]]

Classification Report:
      precision    recall  f1-score   support

setosa      1.00      1.00      1.00        10
versicolor  1.00      1.00      1.00         9
virginica    1.00      1.00      1.00        11

accuracy          1.00        30
macro avg      1.00      1.00      1.00        30
```